

Exercise 4.3

Q.1 Find HCF by factorization method.

(i) $21x^2y, 35xy^2$

Solution

$$21x^2y, 35xy^2$$

Factorization

$$21x^2y = 3 \times 7 \times x \times x \times y$$

$$35xy^2 = 5 \times 7 \times x \times y \times y$$

Common factors = 7, x, y

H.C.F = $7xy$ (product of common factors)

(ii) $4x^2 - 9y^2, 2x^2 - 3xy$

Solution:

$$4x^2 - 9y^2, 2x^2 - 3xy$$

Factorization

$$4x^2 - 9y^2 = (2x)^2 - (3y)^2$$

$$\because a^2 - b^2 = (a+b)(a-b)$$

$$= (2x+3y)(2x-3y)$$

Now,

$$2x^2 - 3xy = x(2x-3y)$$

Common factor = $(2x-3y)$

H.C.F = $(2x-3y)$

(iii) $x^3 - 1, x^2 + x + 1$

Solution:

$$x^3 - 1, x^2 + x + 1$$

Factorization

$$x^3 - 1 = (x)^3 - (1)^3$$

$$\because a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$= (x-1) [(x)^2 + (x)(1) + (1)^2]$$

$$= (x-1) (x^2 + x + 1)$$

Now,

$$x^2 + x + 1 = (x^2 + x + 1)$$

$$\text{Common factor} = (x^2 + x + 1)$$

$$\text{H.C.F} = (x^2 + x + 1)$$

(iv) $a^3 + 2a^2 - 3a, 2a^3 + 5a^2 - 3a$

Solution:

$$a^3 + 2a^2 - 3a, 2a^3 + 5a^2 - 3a$$

Factorization

$$a^3 + 2a^2 - 3a = a(a^2 + 2a - 3)$$

$$= a[a^2 + 3a - a - 3]$$

$$= a[a(a+3) - 1(a+3)]$$

$$= a(a+3)(a-1)$$

$$\text{Now, } 2a^3 + 5a^2 - 3a = a(2a^2 + 5a - 3)$$

$$= a[2a^2 + 6a - a - 3]$$

$$= a[2a(a+3) - 1(a+3)]$$

$$= a(a+3)(2a-1)$$

Common factors = $a(a+3)$

$$\text{H.C.F} = a(a+3)$$

(v) $t^2 + 3t - 4, t^2 + 5t + 4, t^2 - 1$

Solution: (Correction)

$$t^2 + 3t + 4, t^2 + 5t + 4, t^2 - 1$$

Factorization

$$t^2 + 3t + 4 = t^2 + 4t - t + 4$$

$$= t(t+4) - 1(t+4)$$

$$= (t+4)(t-1)$$

$$t^2 + 5t + 4 = t^2 + 4t + t + 4$$

$$= t(t+4) + 1(t+4)$$

$$[\because a^2 - b^2 = (a+b)(a-b)]$$

$$= (t+4)(t+1)$$

$$t^2 - 1 = (t)^2 - (1)^2$$

$$= (t+1)(t-1)$$

Common factor = $(t+1)$

$$\text{H.C.F} = (t+1)$$

(vi) $x^2 + 15x + 56, x^2 + 5x - 24, x^2 + 8x$

Solution:

$$x^2 + 15x + 56, x^2 + 5x - 24, x^2 + 8x$$

Factorization

$$\begin{aligned} x^2 + 15x + 56 &= x^2 + 7x + 8x + 56 \\ &= x(x+7) + 8(x+7) \\ &= (x+7)(x+8) \end{aligned}$$

$$\begin{aligned} x^2 + 5x - 24 &= x^2 + 8x - 3x - 24 \\ &= x(x+8) - 3(x+8) \end{aligned}$$

$$\text{Now, } x^2 + 8x = x(x+8)$$

$$\text{Common factor} = (x+8)$$

$$\text{H.C.F} = (x+8)$$

Q.2 Find HCF of the following

expressions by using division method:

(i) $27x^3 + 9x^2 - 3x - 10, 3x - 2$ (correction)

Solution:

$$27x^3 + 9x^2 - 3x - 10, 3x - 2$$

H.C.F by Division Method

$$\begin{array}{r} 9x^2 + 9x + 5 \\ 3x - 2 \overline{) 27x^3 + 9x^2 - 3x - 10} \\ \underline{\oplus 27x^3 \ominus 18x^2} \\ 27x^2 - 3x - 10 \\ \underline{\oplus 27x^2 \ominus 18x} \\ 15x - 10 \\ \underline{\ominus 15x \ominus 10} \\ 0 \end{array}$$

$$\text{Thus, H.C.F} = 3x - 2$$

(ii) $x^3 - 9x^2 + 21x - 9, x^2 - 4x + 3$

Solution:

$$x^3 - 9x^2 + 21x - 9, x^2 - 4x + 3$$

H.C.F by Division Method

$$\begin{array}{r} x - 5 \\ x^2 - 4x + 3 \overline{) x^3 - 9x^2 + 21x - 9} \\ \underline{- x^3 + 4x^2 - 3x} \\ -5x^2 + 18x - 9 \\ \underline{-5x^2 + 20x - 15} \\ -2x + 6 \\ \underline{-2(x-3)} \\ 0 \end{array}$$

$$\begin{array}{r} x - 1 \\ x - 3 \overline{) x^2 - 4x + 3} \\ \underline{- x^2 + 3x} \\ -x + 3 \\ \underline{-x + 3} \\ 0 \end{array}$$

$$\text{H.C.F} = x - 3$$

(iii) $2x^3 + 2x^2 + 2x + 2, 6x^3 + 12x^2 + 6x + 12$

Solution:

$$2x^3 + 2x^2 + 2x + 2, 6x^3 + 12x^2 + 6x + 12$$

H.C.F by Division Method

$$\begin{array}{r} 3 \\ 2x^3 + 2x^2 + 2x + 2 \overline{) 6x^3 + 12x^2 + 6x + 12} \\ \underline{\oplus 6x^3 \oplus 6x^2 \oplus 6x \oplus 6} \\ 6x^2 + 6 \\ 3(2x^2 + 2) \end{array}$$

$$\begin{array}{r} x + 1 \\ 2x^2 + 2 \overline{) 2x^3 + 2x^2 + 2x + 2} \\ \underline{\oplus 2x^3 \oplus 2x^2} \\ 2x^2 + 2 \\ \underline{\oplus 2x^2 \oplus 2} \\ 0 \end{array}$$

$$\text{Thus H.C.F} = 2x^2 + 2 = 2(x^2 + 1)$$

(iv) $2x^3 - 4x^2 + 6x, x^3 - 2x, 3x^2 - 6x$

Solution:

$$2x^3 - 4x^2 + 6x, x^3 - 2x, 3x^2 - 6x$$

First we find H.C.F of $x^3 - 2x, 3x^2 - 6x$

$$\begin{array}{r} (x^2 - x) \\ (x^2 - x) \overline{) 3x^2 - 6x} \\ \underline{\oplus 3x^2 \ominus 3x} \\ -3x \end{array}$$

(Taking 6 as common) $6x^2 - 6x$

$$\begin{array}{r} (x^2 - x) \\ (x^2 - x) \overline{) 3x^2 - 6x} \\ \underline{\oplus 3x^2 \ominus 3x} \\ -3x \end{array}$$

($\because -3$ is not a common factor, we ignore it)

$$\begin{array}{r}
 x-1 \\
 x \overline{) x^2 - x} \\
 \underline{\oplus x^2} \\
 -x \\
 \underline{\oplus x} \\
 0
 \end{array}$$

Thus H.C.F is "x".

Q.3 Find LCM of the following expressions by using prime factorization method.

(i) $2a^2b$, $4ab^2$, $6ab$

Solution:

$$2a^2b, 4ab^2, 6ab$$

$$2a^2b = 2 \times a \times a \times b$$

$$4ab^2 = 2 \times 2 \times a \times b \times b$$

$$6ab = 2 \times 3 \times a \times b$$

Product of common factors = $2ab$

Product of non-common factors

$$= (a)(2b)(3) = 6ab$$

$$\text{L.C.M} = (2ab)(6ab) = 12a^2b^2$$

(ii) $x^2 + x$, $x^3 + x^2$

Solution:

$$x^2 + x, x^3 + x^2$$

Factorization

$$x^2 + x = x(x+1)$$

$$x^3 + x^2 = x^2(x+1)$$

$$= x \cdot x(x+1)$$

Product of common factors = $x(x+1)$

Product of non-common factors = x

$$\text{L.C.M} = x(x+1) \times x$$

$$\text{L.C.M} = x^2(x+1)$$

(iii) $a^2 - 4a + 4$, $a^2 - 2a$

Solution:

$$a^2 - 4a + 4, a^2 - 2a$$

Factorization

$$a^2 - 4a + 4 = (a)^2 - 2(a)(2) + (2)^2$$

$$= (a-2)^2$$

$$= (a-2)(a-2)$$

$$a^2 - 2a = a(a-2)$$

Product of common factors = $(a-2)$

Product of non-common factors = $a(a-2)$

$$\text{L.C.M} = (a-2) \cdot a(a-2)$$

$$= a(a-2)^2$$

(iv) $x^4 - 16$, $x^3 - 4x$

Solution:

$$x^4 - 16, x^3 - 4x$$

$$x^4 - 16 = (x^2)^2 - (4)^2$$

$$= (x^2 + 4)(x^2 - 4)$$

$$= (x^2 + 4)[x^2 - 2^2]$$

$$= (x^2 + 4)(x+2)(x-2)$$

$$\text{Now, } x^3 - 4x = x(x^2 - 4) = x(x^2 - 2^2)$$

$$= x(x+2)(x-2)$$

$$\text{L.C.M} = (x+2)(x-2)x(x^2+4)$$

$$= (x^2+2)(x-2)x(x^2+4)$$

$$= x(x^2-2^2)(x^2+4)$$

$$= x(x^2-4)(x^2+4)$$

$$= x(x^2)^2 - (4)^2$$

$$= x(x^4 - 16)$$

(v) $16 - 4x^2$, $x^2 + x - 6$, $4 - x^2$

Solution:

$$16 - 4x^2, x^2 + x - 6, 4 - x^2$$

Factorization

$$16 - 4x^2 = 4(4 - x^2)$$

$$= 4[2^2 - x^2]$$

$$= 4(2+x)(2-x)$$

$$= 4(x+2)(-1)(-2+x)$$

$$= (-1)(4)(x+2)(x-2)$$

$$= -4(x+2)(x-2)$$

$$x^2 + x - 6 = x^2 + 3x - 2x - 6$$

$$= x(x+3) - 2(x+3)$$

$$= (x+3)(x-2)$$

$$4 - x^2 = (2)^2 - (x)^2$$

$$= (2+x)(2-x)$$

$$= (x+2)(-1)(-2+x)$$

$$= -1(x+2)(x-2)$$

Product of common factors

$$(-1)(x+2)(x-2)$$

$$= (-1)(x^2 - 4) = (4 - x^2)$$

Product of non-common factors = $4(x+3)$

$$\text{L.C.M} = 4(4 - x^2)(x+3)$$

Q.4 The HCF of two polynomials is $y-7$ and their LCM is $y^3 - 10y^2 + 11y + 70$. If one of the polynomials is $y^2 - 5y + 14$, find the other.

Solution:

$$\text{H.C.F} = y-7$$

$$\text{L.C.M} = y^3 - 10y^2 + 11y + 70$$

$$p(y) = y^2 - 5y - 14$$

We know that

$$p(y) \cdot q(y) = \text{L.C.M} \times \text{H.C.F}$$

$$q(y) = \frac{\text{L.C.M} \times \text{H.C.F}}{p(y)}$$

$$q(y) = \frac{(y^3 - 10y^2 + 11y + 70) \times (y - 7)}{y^2 - 5y - 14}$$

$$q(y) = (y - 5)(y - 7)$$

$$q(y) = y^2 - 12y + 35$$

Q.5 The LCM and HCF of two polynomial $p(x)$ and $q(x)$ are $36x^3(x+a)(x^3-a^3)$ and $x^2(x-a)$ respectively.

If $p(x) = 4x^2(x^2 - a^2)$, find $q(x)$.

Solution:

$$\text{L.C.M} = 36x^3(x+a)(x^3-a^3)$$

$$\text{H.C.F} = x^2(x-a)$$

$$P(x) = 4x^2(x^2-a^2)$$

We know that

$$p(x) \cdot q(x) = \text{L.C.M} \times \text{H.C.F}$$

$$q(x) = \frac{\text{L.C.M} \times \text{H.C.F}}{p(x)}$$

$$q(x) = \frac{36x^3(x+a)(x^3-a^3) \times x^2(x-a)}{4x^2(x^2-a^2)}$$

$$q(x) = \frac{9x^3(x^3-a^3) \cdot x^2(x^2-a^2)}{x^2(x^2-a^2)}$$

$$q(x) = 9x^3(x^3-a^3)$$

Q.6 The HCF and LCM of two polynomials is $(x+a)$ and $12x^2(x+a)(x^2-a^2)$ respectively. Find the product of the two polynomials.

Solution:

$$\text{H.C.F} = (x+a)$$

$$\text{L.C.M} = 12x^2(x+a)(x^2-a^2)$$

Let $p(x)$ and $q(x)$ be required polynomials

We know that

$$p(x) \times q(x) = \text{L.C.M} \times \text{H.C.F}$$

$$p(x) \times q(x) = 12x^2(x+a)(x^2-a^2) \times (x+a)$$

$$= 12x^2(x+a)^2(x-a) \times (x-a)$$

$$= 12x^2(x+a)^3(x-a)$$